

RENANCE

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

MTH 413 — Past Questions & Materials
by Renance

Course: MTH 413
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 2
Levels: 400

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Paper 1 of 2 — MTH 413 General Topology First Semester 2018 2019

Level 400 • First Semester • PastQuestion

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RAT. UNIVERSITY OF PETROLEUM RESOURCES. EPFURUN

Petlad COLLEGE. OF SCIENCE

DEPARTMENT OF MATHEMATICS AND COMPUTER SCIENCE

FIRST SEMESTER EXAMINATION 2018/2019

Course Code/Title: MTH 413 -

Instructions: Attempt any 4

General Topology

questions Time Allowed: 2 hours

Question 1 (10 Marks)

- Let $X = (a, b, c, d)$. Show that $\tau = \{\emptyset, \{c, d\}, \{a, c, d\}, \{b, c, d, e\}, X\}$ is a topology on X . (7 marks)
- Let (A, τ) be a subspace where $A = (a, b, c)$. Compute τ , and show that τ is a topology on A . (6 marks)
- Find the accumulation point and the closure of A . (4 marks)

Question 2 (14 Marks)

- When is a space said to be (i) Metrizable (ii) Hausdorff (iii) T_1 Space (iv) T_2 Space. — (8 marks)
- Show that every Metric space is Hausdorff. (2 marks)
- Justify that a topological space is T_2 space if and only if it is closed. (4 marks)

Question 3 (17 Marks)

- When is the function $f: (X, \tau) \rightarrow (Y, \sigma)$ said to be continuous? (3 marks)
- If (X, D) is a discrete space, then every function $X \rightarrow Y$ is continuous. — (6 marks)
- Let $X = (a, b)$ with $\tau = \{\emptyset, \{b\}, X\}$ and $Y = (c, d, f)$ with $\sigma = \{\emptyset, \{c\}, \{A\}, \{d, f\}, Y\}$. Define $f: Y \rightarrow X$ by the maps: $f(d, b) = a$ and $f(a, d, f) = b$ respectively. Show that $f: Y \rightarrow X$ is continuous. (8 marks)

Question 4 (17 Marks)

- When is a function $f: (X, \tau) \rightarrow (Y, \sigma)$ said to be a Homeomorphism? (3 marks)
- Let $f: [0, 1] \rightarrow [a, b]$ be a map defined by $f(x) = a(d - x) + bx$. Show that $[0, 1]$ and $[a, b]$ are Homeomorphic. (6 marks)
- Explain with clear examples the following terms (1) Sierpinski space (ii) Finer topology (iii) Coarser topology (3 marks)

Question 5 (14 Marks)

- When is a topological space said to be (1) connected (2) compact? (2 marks)
- State without proof the Heine-Borel Theorem. (2 marks)
- Show whether or not the class $\mathcal{P} = \{A_n : n \in \mathbb{N}\}$ is an open cover of $X = (0, 1)$ where $A_n = (0, \frac{1}{n})$. (2 marks)

compact. (7's marks)

Question 6 (17% Marks)

u Whee IS a lis of subset ol T rial la be a basis for Cunarks)

bh Show whether or not $A = \{ (a, b, \phi), (c, d), (d, e) \}$ ds a basis fora topology on the set $X =$

(a, b, c, d, e) (?niarks)

© find the Lopalow) on $A = \{ (c, b, \phi, c) \}$ venerated b, $A = \{ (a, 0, c^*), \phi, (, dl), (dL, @) \}$. (7 '> marks)

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Paper 2 of 2 — MTH 413 General Topology 2020 2021 First Semester

Level 400 • First Semester • PastQuestion

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Marking Guide



FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN
COLLEGE OF SCIENCE
DEPARTMENT OF MATHEMATICS
FIRST SEMESTER EXAMINATION 2020/2021

Course Code/ Title: MTH 413-General Topology

Instructions: Attempt any 4 questions. Time Allowed: $2\frac{1}{2}$ Hours

Question 1:

- What is a topology and a topological space? Let $X = \{a, b, c\}$, define six topologies on X such that each consists of four open sets. (6 marks).
- Let T be the class of subsets of $\mathbb{N}$ consisting of $\emptyset$ and all subsets of $\mathbb{N}$ of the form $E_n = \{n, n+1, n+2, \dots\}$ with $n \in \mathbb{N}$ (i) Show that T is a topology on $\mathbb{N}$ (ii) List the open sets containing the positive integer 6 and show that T_6 with $\emptyset$ inclusive is again a topology on $\mathbb{N}$ (6 marks).
- Define (i) an accumulation point (limit point) (ii) the derived set. Let $(\mathbb{N}, T)$ be a topological space as defined in (b) above. Then, outline the accumulation points of the set $A = \{4, 13, 28, 37\}$ and hence find the derived set of A (A'). ($5\frac{1}{2}$ marks).

Question 2:

- Let X be a topological space and $A \subset X$. Then define (i) closure of A (ii) Given that $B \subset X$ when is A said to be dense in B ? (4 marks).
- Let A be a subset of a topological space X . Then prove that the closure is the union of A and its set of accumulation points. i.e. $\bar{A} = A \cup A'$ (6 marks).
- Let

$$T = \{X, \emptyset, \{a\}, \{a, c, d\}, \{a, b, c, d\}, \{a, b, e\}\}$$
 be a topology on $X = \{a, b, c, d, e\}$. Then
 - List the closed subsets of X
 - Determine the closure of the sets $\{a\}$, $\{b\}$ and $\{c, e\}$
 - Which sets in (ii) above are dense in X ?
 - If $A = \{b, d, e\}$ find the closure point of A . ($7\frac{1}{2}$ marks).

Question 3:

- State the Kuratowsky's axioms of closure ($4\frac{1}{2}$ marks).
- Let X be a topological space, and A, B subsets of X . Prove that (i) if $A \subset B$ then $\bar{A} \subset \bar{B}$. (ii) $\overline{A \cup B} = \bar{A} \cup \bar{B}$ Using these notions, show that the Kuratowsky's axioms hold (7 marks)